

YUE WU

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EDUCATION

Southern University of Science and Technology, Shenzhen, Guangdong 08/2021 – 06/2025

Bachelor of Statistics, GPA: 3.80/4.00, Rank: 6/36

- Main Courses: Statistical Learning(98), Time Series Analysis(92), Operations Research and Optimization(90), Probability Theory and Mathematical Statistics(97), Statistical Computation and Software(91)

PUBLICATION

[1] **Wu, Y.**, Li, Y., & Yang, P. (2024). SimLOB: Learning Representations of Limited Order Book for Financial Market Simulation. *IEEE Transactions on Artificial Intelligence*. (Under Review)

[2] **Wu, Y.**, Liao, Y. et al. (2024). An Automatic and Speech-based Cross-Lingual Classification Framework for Early Screening of Cognitive Impairment. *Alzheimer's & Dementia: The Journal of the Alzheimer's Association*. (Under Review)

RESEARCH EXPERIENCE

Personalized Knowledge Graph Summarization with LLM **Research Assistant**
Supervisor: Prof. Yujun Yan, Dartmouth **08/2024-12/2024**

- Goal: Designed a personalized knowledge graph summarization algorithm with LLM based on historical data.
- Contribution: Conducted research on Knowledge Graph Reasoning algorithms and Question-Answering databases. Utilized language model to construct a text-based question similarity graph and built historical data based on it. Designed the algorithm using LLM as logical judgment and summarization for the final personalized knowledge graph.

Representation Learning for Multi-Agent System Calibration **Research Assistant**
Supervisor: Prof. Peng Yang, SUSTech **07/2023-05/2024**

- Contribution: Introduced representation learning for calibration in this field and constructed an Autoencoder based on Transformer to represent multi-feature LOB (Limit Order Book) data. The data was mapped to the latent space using a pre-trained Encoder for model calibration, resulting in over a 50% performance improvement. Designed the experimental process and model training and contributed to the paper writing.
- Co-authored a paper submitted to TAI. [Preprint Link Here](#).

Alzheimer Diagnosis through Audio Analysis **Research Assistant**
Supervisor: Prof. Xiang Fan, Peking University Hospital Shenzhen **08/2024-11/2024**

- Model Building: Developed a model framework for early cognitive impairment screening based on a multilingual picture description task. This framework addresses the challenge of integrating multilingual data and focuses on mild cognitive impairment, offering strong practical value and achieving strong performance: AUC=0.75, ACC=0.77. I built the entire model: ASR for speech-to-text conversion, a language model for translation alignment and feature extraction, and a fine-tuned neural network as the classifier.
- Write a paper as the first author, submitted to ADJ. [Manuscript Link Here](#).

INTERNSHIP EXPERIENCE

ASR & Audio Data Processing Intern **Internship**
Supervised: Shiyin Kang, SenseTime Research Institute **02/2025-05/2025**

- Built and optimized a GPU inference pipeline for SenseTime ASR and SenseVoice ASR models; redesigned dataset and processor modules to support multi-format audio data and record detailed runtime metrics for each module. Improved processing efficiency by 50% and resolved issues like truncated endings and incorrect English transcription.
- Designed a communication dialogue selection pipeline, using VAD, ASR, and other models to extract quality metrics, speaker timelines, and transcriptions. Selected high-quality conversations based on pause duration, SNR, OVRL, AST, etc.; visualized results and accelerated the filtering process through algorithmic optimization.
- Trained a BERT-based NER model to extract artist and song names from music filenames, combining it with embedding-based filename similarity for duplicate detection. Developed a structured and searchable metadata database.

Leveraging Synthetic Data Inducing LLM Capabilities

Intern Team Leader

Supervised: Yipeng Cheng, xDAN AI

03/2024-06/2024

- Data Augmentation: Expanded over 10 classical datasets through data augmentation and diversification, guided by WizardLM, DEITA, and other papers, utilizing advanced tools such as Distilable and DataDreamer. Quantitatively evaluated LLM data based on complexity and quality using UltraFeedBack. Used synthetic data to induce specific capabilities.
- Quantitative Scoring: Based on papers like DEITA, developed a framework using large language models for sample comparison to assign quality scores. The framework quantitatively analyzes text from perspectives such as complexity and relevance and, combined with data augmentation, constructs a high-quality dataset.

PROJECT EXPERIENCE

Analysis of Bilibili Content Creators

Course Project(Best)

Supervised: Prof. Zeng Li, Southern University of Science and Technology

11/2023-12/2023

- EDA: Applied ANOVA, two-way ANOVA, and ANCOVA to assess the impact of tags, gender, video length, and their interactions on follower count. Utilized GMM fitting to classify video lengths on Bilibili and explored their relationship with follower count.
- Feature Selection: Performed variable selection using RandomForest, Lasso, XGBoost, and DCSIS to identify critical features influencing follower count.
- Regression& Clustering: Conducted regression analysis with weighted least squares, LightGBM, and five other models; provided model interpretations. Employed HDBSCAN for clustering content creators and developed an underestimation mechanism to identify promising categories and video formats, offering recommendations for content creator positioning.

EXTRA-CURRICULAR ACTIVITIES

Organizing and Participating in a Self-Supervised Learning Seminar

6/2023-8/2023

- Based on A Cookbook of Self-Supervised Learning, we organized a seminar to about self-supervised learning and meta-learning. I studied relevant papers and conducted multiple knowledge-sharing sessions.
- Seminar Link

Organizing and Participating in a Graph Neural Networks Seminar

3/2024-4/2024

- I organized a seminar focused on Graph Learning, based on SC224W and Graph Representation Learning; we extensively studied GNN, Knowledge Graph, Graph Reasoning, Graph Recommendation and its applications, read several papers, and learned Pytorch_Geometric.

Chairperson of the Student Union of Zhiren (Highest-level Position)

9/2023-9/2024

- Collaborated with multiple student organizations to organize dozens of campus-level events and received the First Prize for Outstanding Student Organization (Highest Rating) during the tenure.
- Participated in the filming and design of the college documentary.

Captain of the University Orienteering Team and the Orienteering Club

9/2022-9/2024

- Organized multiple campus-level events to promote orienteering, and the club received the Highest Rating).
- I won third place in the 2023 Orienteering and fourth place in the 2022 Orienteering at the Guangdong University Games.

AWARD

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| • The Grand Prize of School Motto "Innovation" Series Scholarship (1%,Top 9, SUSTech) | 2024 |
| • First Class of the Merit Student Scholarship(1%, SUSTech) | 2024 |
| • iGEM International Competition Gold Medal(10%, Global) | 2023 |
| • First Class (First Place) of the Merit Student Scholarship(1%, SUSTech) | 2023 |
| • First Prize in the China Undergraduate Mathematical Contest in Modelling(10%, Guangdong) | 2023 |
| • Student Cadre Model of Excellence(1%, SUSTech) | 2023 |

SKILLS & LANGUAGES

Computer Skills: Python,Java,C++, R,SAS,Hadoop,Spark,Pytorch,Linux,Latex

Language:Native in Chinese,Proficient in English(TOEFL 103).